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Decentralized Health Care Management System with Ai Diagnostic

Connecting Patients to the right doctor with AI + real-time chat

Team : Nemesis

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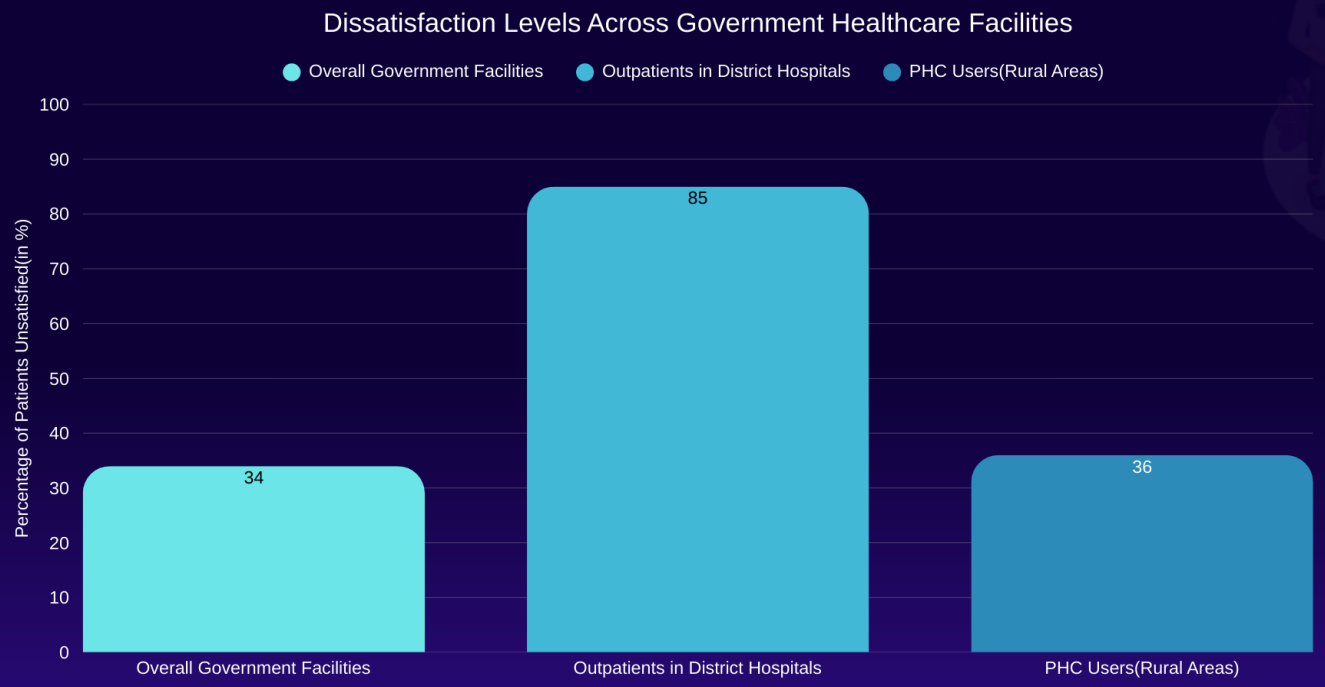
01

PROBLEM STATEMENT

Patients often struggle to find the right doctor quickly, as current platforms lack AI-based guidance, smart filtering, secure data handling, and seamless payment or real-time consultation options— leading to delays, confusion, and privacy risks.

WHY IT MATTERS ?



Source:Institute of Social and Economic Change(ISEC),July 2023.Additional study on rural PHCs also cited

PROBLEM → AIM

In today's healthcare system, **Renowned Hospitals** dominate the market because of their visibility, resources, and reputation.

This often results in:

High consultation costs for patients.
Limited access for people in smaller towns or rural areas.
Squanders of local doctors and clinics, who may be equally skilled but lack socialization.

Our project aims to **enhance and improve their potential** by creating a **decentralized healthcare platform** where:

1. **Patients** can directly connect with qualified doctors, not just those from famed hospitals.
2. **AI-driven specialization matching** helps patients find the right doctor based on their symptoms, instead of reverting to famed hospitals.
3. **Ratings, proximity, and specialization filters** allow patients to choose **trusted and nearby doctors**, making healthcare **more accessible and affordable**.
4. **Decentralized medical data** ensures **patients own their health records**, not big hospital chains.

OUR SOLUTION

1. AI-powered specialization suggestion (Gemini API)
2. Smart doctor discovery (rating, distance, specialization) with AI voice assistance feature.
3. Secure payments via Cashfree API
4. Real-time doctor-patient chat and Doctor-controlled sessions
5. File transfer & Navigational Aid Features

TECH STACK

Frontend

React.js → For building dynamic and interactive user interface

Tailwind CSS → For modern, responsive styling and UI components

Axios → For making API calls to backend and external services

Backend

Node.js → Runtime environment for server-side logic

Express.js → Web framework for handling routes and APIs

Socket.io → For real-time communication (doctor–patient chat)

CORS → Middleware for secure cross-origin requests

Database

MongoDB → NoSQL database for storing user and doctor data

Mongoose → ODM (Object Data Modeling) library for MongoDB, to manage schemas and queries

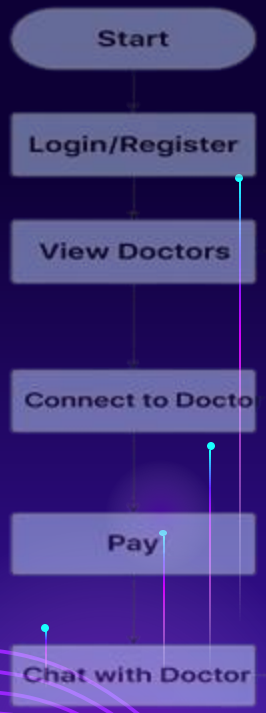
APIs & Integrations

Gemini API → Generative AI for analyzing patient symptoms and suggesting specialization

Cashfree API → Secure payment gateway integration for consultations

USER JOURNEY

- **1**. User Login/Register
- **2**. Doctor Discovery (filters)
- **3**. AI suggests specialization along with voice assistance feature.
- **4**. User selects doctor
- **5**. Payment via Cashfree
- **6**. Chat opens
- **7**. Patient can consult to doctor
- **8**. Doctor ends chat



IMPACT AND BENEFITS

⚡ **Faster Access to Healthcare AI** instantly guides patients to the right specialist, reducing delays and misdiagnosis.

🔐 **Enhanced Data Privacy & Security Decentralized** storage ensures medical records are owned and controlled by patients, reducing ata leaks.

📍**Location-Aware** Doctor DiscoveryUsers can quickly find nearby doctors using real-time location and distance filtering (Haversine), improving accessibility.

💬 **Seamless Doctor–Patient Communication Integrated** payment + real-time chat ensures smooth consultations and guarantees doctors are paid fairly.

📈 **Improved Healthcare Efficiency** Reduces hospital load, organizes doctor schedules, and streamlines the entire consultation process.

Future Scope

Smarter AI Diagnostics – Upgrade to full medical AI that analyzes reports, images, and vitals to predict diseases and urgency levels.

Decentralized Health Records – Store encrypted medical data on blockchain/IPFS so patients fully own and control access to their records.

Telemedicine & Mobile App – Launch mobile apps with video consultation, voice calls, and e-prescriptions for seamless care.

Insurance & Pharmacy Integration – Enable instant insurance claims and doorstep medicine delivery from nearby pharmacies.

Doctor Analytics & Personal Health Assistant – Use AI to track doctor performance, patient outcomes, and build a smart health companion for users.

THANK YOU

TEAM : NEMESIS